

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

C05: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Technical Presentation

Code of Conduct:

I _FURTHOSE SAMREEN (192511164)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: FURTHOSE SAMREEN B

Technical Presentation

1. Reactive Agent-Based Traffic Control System

Prepare a technical presentation on designing a **Reactive Agent** for a smart traffic control system. Explain how the agent responds to real-time traffic inputs, manages signal changes, and ensures minimal congestion using streaming data processing.

2. Goal-Based Agent for Healthcare Diagnosis

Develop a presentation on a **Goal-Based Agent** used in an AI healthcare system. Describe how the agent sets diagnostic goals, evaluates patient data, and selects optimal treatment pathways using decision-making logic and data pipelines.

3. Utility-Based Agent for E-Commerce Recommendations

Create a technical presentation on a **Utility-Based Agent** that generates personalized product recommendations. Explain how the agent assigns utility values to different options and optimizes user satisfaction using large-scale data processing.

4. Learning Agent for Personalized Education System

Present the design of a **Learning Agent** in an AI-driven educational platform. Explain how the agent improves performance over time using feedback, adapts content difficulty, and integrates with distributed data processing systems like PySpark.

5. Multi-Agent System for Disaster Management

Prepare a presentation on a **Multi-Agent System** where multiple agents (sensor agents, decision agents, communication agents) collaborate for disaster detection and response. Explain coordination strategies, data sharing, and real-time decision-making.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

1. Reactive Agent-Based Traffic Control System

A Reactive Agent responds directly to changes in its environment. In a smart traffic system, it continuously monitors traffic conditions and changes traffic signals accordingly.

Problem: Fixed-time signals can cause unnecessary waiting and congestion when traffic density changes.

Objectives:

- Monitor traffic in real time.
- Detect traffic density.
- Dynamically control traffic signals.
- Reduce congestion and waiting **time**.

Architecture:

Traffic Sensors/Cameras → Streaming Data → Reactive Agent → Traffic Analysis → Signal Controller → Traffic Signals

Working:

1. Sensors and cameras collect vehicle-density information.
2. Streaming data is continuously processed.
3. The reactive agent checks the current traffic condition.
4. If traffic is heavy on one road, green-signal time is increased.
5. When traffic decreases, signal timing is adjusted.

Decision:

- High traffic → Increase green time.
- Medium traffic → Normal signal time.
- Low traffic → Reduce green time.

Role of PySpark: PySpark processes large amounts of streaming traffic data in a distributed manner and analyzes information from multiple roads simultaneously.

Advantages:

- Reduces congestion.
- Provides real-time response.
- Improves road utilization.
- Reduces waiting time.

Conclusion: A reactive-agent traffic system continuously responds to real-time traffic inputs and dynamically manages signals to minimize congestion.

2. Goal-Based Agent for Healthcare Diagnosis

A Goal-Based Agent makes decisions by considering a specific goal. In healthcare, the goal is to identify a suitable diagnosis and treatment pathway using patient information.

Architecture:

Patient Data → Data Pipeline → Goal-Based Agent → Diagnosis Analysis → Decision Making → Treatment Pathway

Working:

1. Patient information is collected.
2. Data is processed through a data pipeline.
3. The agent establishes a diagnostic goal.
4. Symptoms and patient information are evaluated.
5. Different possible decisions are compared.
6. The best treatment pathway is selected.

Decision Logic:

Patient Data → Possible Diagnoses → Goal Evaluation → Best Decision → Treatment Recommendation

Role of Data Processing: Distributed processing can handle large healthcare datasets efficiently.

Advantages:

- Goal-oriented decision making.
- Faster patient-data analysis.
- Supports healthcare professionals.
- Handles large datasets.

Conclusion: A goal-based agent uses patient data and diagnostic goals to select an appropriate healthcare pathway.

3. Utility-Based Agent for E-Commerce Recommendations

A Utility-Based Agent compares different choices based on their usefulness or utility value. In e-commerce, it recommends products that maximize customer satisfaction.

Architecture:

User Data → PySpark Processing → Product Analysis → Utility Calculation → Utility Agent → Personalized Recommendations

Working:

1. User searches, purchases, ratings, and preferences are collected.
2. Large-scale user and product data is processed.
3. Utility values are calculated for different products.
4. Products are compared.
5. Products with higher utility are recommended.

Example: For laptops, the agent can consider:

- Price
- Rating
- Features
- Previous user preferences

The system calculates overall utility and ranks suitable laptops.

Role of PySpark: PySpark processes large customer and product datasets in parallel.

Advantages:

- Personalized recommendations.
- Improved customer satisfaction.
- Efficient large-scale processing.
- Better product selection.

Conclusion: A utility-based agent improves e-commerce recommendations by assigning utility values and selecting products that maximize user satisfaction.

4. Learning Agent for Personalized Education

A Learning Agent improves its performance using feedback and previous experience. In education, it adapts learning content according to student performance.

Architecture:

Student Activity → PySpark → Performance Analysis → Learning Agent → Content Adaptation → Student Feedback → Learning

Working:

1. Student activities and performance data are collected.
2. PySpark processes learning data.
3. The agent analyzes student performance.
4. Feedback is used to evaluate the current content.
5. Content difficulty or type is changed.
6. The agent continues learning and improves over time.

Learning Process:

Student Performance → Feedback → Agent Learning → Content Adjustment → Improved Learning

Role of PySpark: It efficiently processes distributed data from many students.

Advantages:

- Personalized learning.
- Adaptive difficulty.
- Continuous improvement.
- Real-time feedback.
- Supports many students.

Conclusion: A learning agent provides personalized education by learning from feedback and adapting content difficulty.

5. Multi-Agent System for Disaster Management

A Multi-Agent System (MAS) consists of multiple intelligent agents that cooperate to solve complex problems. In disaster management, different agents monitor, analyze, communicate, and coordinate emergency responses.

Architecture:

Sensors → Sensor Agents → Decision Agent → Communication Agent → Emergency Response

Types of Agents:

1. Sensor Agent – Collects environmental and disaster-related information.
2. Decision Agent – Analyzes information and determines the appropriate response.
3. Communication Agent – Shares warnings and decisions with authorities and emergency teams.

Working:

1. Sensor agents continuously collect data.

2. Information is shared with decision agents.
3. Decision agents analyze the situation.
4. Communication agents distribute alerts.
5. Agents coordinate their actions for disaster response.

Coordination:

Data Collection → Data Sharing → Decision Making → Alert → Response

Role of PySpark: Distributed processing such as PySpark can analyze large amounts of disaster data from multiple sources efficiently.

Advantages:

- Fast disaster detection.
- Real-time decision-making.
- Effective communication.
- Cooperation between agents.
- Better emergency response.

Conclusion: A multi-agent system improves disaster management by allowing sensor, decision, and communication agents to collaborate effectively.